

Reformulating the comfort-zone measurement inside active inference, a design note

2026-09-18, written after cards JJ.2, JJ.3, JJ.2b, RE.1 and RE.2, on Jonas's question: "think more about how the problem we have possibly could be reformulated to fit in the active inference framework. What would we have to do differently for this to work. Again, I feel that with the right formulation and the right framing, it should work."

Status: a design note. Nothing here is authorized, nothing has been implemented, and the rules in §6 become binding only when Jonas has read them. Where a value is a placeholder it says so. Opinions are marked as opinions. Every number quoted carries its file.

0 The answer in two paragraphs

The framework is not what failed, and there is now direct evidence for that rather than an argument. Card JJ.2b gave the model its own policy space — the released planner instead of a hand-made longitudinal menu — and the matched-TTC ordering, the thing gate R.2, card JJ.2 and card JJ.3 all failed on, went from **0 of 24 rows** in the human direction to **19 of 24**, with $p(\text{axis, gap})$ turning from +0.848 to -0.308. That is the largest movement any card has produced on this design, and it came from changing the *framing*, not a constant. What it did not do is make the axis usable: the ordering is right-signed and weak, the held-out score stays at chance, and the reason is the second paragraph.

What failed is one identification, made at the very start and never questioned, and card RE.2 has now bounded how much of the failure it accounts for: sweeping 648 parameter vectors over everything a driver could plausibly differ in leaves **0 of 648** ordering even half the matched-TTC rows. So this is not a calibration problem. It is that **the released model's braking-margin term was taken to be the comfort-zone boundary**. `src/aidriver/preferences.py` says so in its own docstring — "that boundary, $a_{\text{ego, req}} = -a_{\text{max}}$, is the model's own operationalisation of a comfort-zone boundary, and `comfortzone` builds on it" — and every card since has measured a level set of it. Card RE.2 now measures what that term does on the human data: its contribution to the model's criticality signal correlates **-0.861** with the share of participants who would intervene. It is not weakly related to comfort; it is **inverted**. Meanwhile the one factor in the released preference that has the shape of a comfort boundary, the inverse-tau preference buried inside the collision factor, correlates **+0.358** — the right sign — and is swamped by four orders of magnitude. The reformulation is therefore not a new construction bolted on top; it is to stop measuring the comfort zone on the dread-zone term, to put the comfort factor on the variable the data name, and to fit the driver's parameters instead of assuming the authors'.

1 What we now know, with the files

finding	number	file
The released criticality signal is Eq. 13's ϵ under the current plan, and our $G(\text{continue})$ is that same object	agree to 7e-16	<code>replication/causation/re1/re1_rear_end_criticality.md</code> §0
ΔG — best alternative minus continue — is a quantity the model never forms	the planner uses only the <code>argmin</code>	ibid.
On a certain constant-speed prediction, ϵ is the same at a 0.5 s headway at 110 km/h, at 3.5 s, and on an empty road	23 nats, all three	ibid. §D
At matched TTC, ϵ is ordered backwards	$p(\epsilon, \text{gap})$ +0.83 to +1.00 against the humans' -0.862	ibid. §B2
The released planner's escape on this design is a steer, never a brake	36 of 36 cells, and 378 of 378 on the study's real traces	ibid. §C; <code>out/jj2b_steer_menu.md</code> §2
Giving the model its own policy space repairs the DIRECTION of the ordering	matched-TTC rows ordered the human way go from 0 of 24 to 19 of 24 ; $p(\text{axis, gap})$ from +0.848 to -0.308	<code>out/jj2b_steer_menu.md</code> §3
...but not the magnitude: the repaired ordering is still weak	$p(\text{axis, share})$ +0.169, held out 0.3161, still chance	ibid.

finding	number	file
On the study's stimuli the speed, acceleration, steering and lane factors contribute exactly zero under "continue"; ϵ is collision + safety and nothing else	$-1.2e-14$ at every one of 378 cells	replication/czb/out/re2_preference_family.md §4
The safety factor alone is inverted against the response	$p(\text{share})$ -0.861 , $p(\text{gap})$ +0.810	ibid.
The collision factor alone has the right sign but a weak ordering	$p(\text{share})$ +0.358 , held out 0.308	ibid.
No parameter vector in the released family orders the cells: 648 swept, the sign is reachable, the ordering is not	306 of 648 give $p(\text{gap}) < 0$, best -0.267; 0 of 648 order more than half the 24 matched-TTC rows; best held out 0.2705	ibid. §1, §5
Most of what the sign responds to is the severity's linearity in closing speed	flattening it moves the median $p(\text{gap})$ from +0.296 to -0.249	ibid. §2
ΔG on the cut-in scores chance	0.3202 against chance 0.320	out/jj2_rollout_cutin.md
The project's own best axis is the optical expansion rate	0.1137 ungated, 0.1027 gated, noise floor 0.118	out/cutin2_gate.md, out/cutin2_looming.md
Drivers are four times sharper in the car than in front of a frozen clip	ratio 4.34 [2.38, 7.93]	out/jj4_precision_spread.md
A bigger apparent size gives a sharper criterion	truck σ_{resp} 0.273 against the car's 0.424, +60.6 LOPO	ibid.

The last two matter more than they look, and §4 item 5 says why.

2 Three assumptions the project made that active inference does not require

2.1 That the comfort-zone boundary is a level set of a scalar. In active inference behavior follows from $\text{argmin}_{\pi} G(\pi)$. There is no scalar called criticality. The line between "comfortable" and "not" is the **switching surface** where the argmin stops being "continue", and that surface is in general not a level set of any of the scalars this project has tried — the pragmatic deficit (gate R.2), ϵ (card RE.1), ΔG (card JJ.2). Forcing a $\Phi((x - c) / \sigma)$ response model onto it assumed a monotone scalar that the framework never promised. *[Opinion: this is the assumption that produced the most wasted work, because it is invisible — it looks like the neutral statistical form the project uses everywhere else.]*

2.2 That the model's parameters are known and the scene is the unknown. The authors calibrate $a_{0V, \min}$ per scenario, through a separate free-following analysis; their parameter vector is a property of their simulation study, not of any driver. The project then asked "does this vector's field predict humans?", which tests someone else's calibration. The native question is the inverse: **a driver's preferences are what a driver IS, so infer them from the responses**. The comfort-zone boundary is then derived from the fitted parameters, per driver, per scenario, by construction.

Card RE.2 tested this assumption on its own and the answer is only half what I expected, which changes item 3 of §4 from a suggestion into a requirement. Sweeping 648 vectors over the five parameters a driver could plausibly differ in, the sign of the ordering is reachable — 306 of 648 give $p(\text{gap}) < 0$ — but **no vector orders even half of the 24 matched-TTC rows** and the best held-out score is 0.2705, chance. So fitting θ instead of assuming the authors' is necessary and **not sufficient**: within this functional form there is no driver whose comfort zone orders these cells. The factor the sign responds to is the collision severity's proportionality to impact speed, which is a property of the form, not a constant a driver could differ in.

2.3 That the response is a function of the true scene. The agent acts on its belief, and the belief depends on the paradigm. A frozen monocular clip conveys distance and closing speed far less precisely than a moving binocular view, and card JJ.4 has now measured exactly that: σ_{resp} 0.858 s on video against 0.198 s on the test track, ratio 4.34. In the current formulation that ratio is a nuisance parameter. In the reformulation it is **the observation model**, and it is measured rather than fitted.

3 The reformulation

Measure the driver's preferences, not the scene's criticality. $P(\text{intervene} \mid \text{scene}, \theta, \text{paradigm}) = P(\text{argmin}_{\pi} G(\pi; \theta, b) \neq \text{continue})$, where **b** is the belief the paradigm affords, **θ** is a small set of the driver's own preference

parameters, and the policy set is the model's own — including steering. Fit θ per driver. The comfort-zone boundary is the switching surface θ induces; the deliverable, a percentile over drivers, is a percentile over θ , expressed in whatever scene coordinate an application wants.

What this buys, and each of these is a failure the current formulation has actually suffered:

- **The axis problem disappears.** No scalar has to be monotone in criticality. Card JJ.2's DROP, card JJ.3's left turn, gate R.2 — all three were failures of monotonicity, not of the framework.
- **The gate disappears.** The pre-onset cells are the cells where continuing wins under any plausible θ , so $P(\text{intervene})$ falls to the lapse rate. Card G.1's fitted gate, card HS.1's surprise gate and card PC.1's projected-conflict gate were all attempts to bolt on by hand what the argmin gives for free.
- **The units problem disappears.** Query **EL.Q4** — "each scenario's axis is in different units, so what does EL.2 divide by?" — has no content once the shared object is θ , which has its own units (m/s^2 , s , rad/s) and is shared across scenarios by construction. EL.2 stops being a rescaling exercise and becomes a transfer test.
- **The trait test becomes a real test.** Card TR.1's +0.647 across mixed units, and card JJ.3's +0.261 on a two-valued axis, were both measuring through a broken instrument.

4 What would have to be done differently, item by item

1. Stop using the braking margin as the boundary. This is the one change that is not optional. $\text{safety_margin} = a_{\text{req}} + a_{\text{max}}$ is the boundary of **physical avoidability**, not of comfort — it asks whether the ego could still avoid a lead braking at $a_{\text{OV,min}}$ after a reaction time. The project's own vocabulary already separates the two: the handbook's *comfort zone* and *dread zone*, and query TT.Q2 parks "the dread boundary" as a separate object. RE.2 measures the cost of having conflated them: $p(\text{share}) = -0.861$. **The braking margin is a good dread-zone boundary and an inverted comfort-zone boundary**, and the project has been fitting comfort levels on it since gate R.1.

2. Put the comfort factor on the variable the data name. The released preference already contains a comfort-shaped factor: inside $\log_collision_pref$, when there is no collision and the other vehicle is ahead, a Gaussian over **inverse tau**, $\tau^{-1} = \varphi/\varphi$, with mean 0.2 s^{-1} and sd 0.125 s^{-1} , one-sided in the released code. This is the only factor in $p(o)$ that bounds an approach *rate* rather than punishing a catastrophe, and it is the only one whose contribution correlates with the human response in the right direction (+0.358). Two candidate one-parameter families, both native to a model that already observes (φ, φ) :

- **τ^{-1} preference, parameter μ_{τ} .** "The inverse TTC I am content with." Already in the code.
- **θ preference, parameter σ_{θ} .** A preference on the optical expansion rate itself. This is the project's own measured axis (card EL.1b: 0.1137 held out against the gap's 0.1522 and the field's 0.3471, at a noise floor of 0.118) with a population already estimated (card EX.2: median 0.034 rad/s, 80th percentile 0.071 rad/s at first exposure).

They differ by a factor of the angular size: $\theta \approx \tau^{-1} \times (W / \text{gap})$. Which one wins is an open empirical question and not a settled one — query DECK.Q1 records that Xue et al. (2018) found τ^{-1} fitting better than θ for both threshold and accumulator models, while this project's own cut-in data prefer θ . **Both are one-parameter families and both should be fitted.**

3. Give the catastrophe factors their proper role. In the released $p(o)$ the collision cost is $-10\,000$ and the road-edge cost $-15\,000$, against a comfort factor of order 1. A model whose comfort term is 10^{-4} of its catastrophe term cannot have its comfort boundary read off the sum: the deposit's own benign-following decomposition puts the collision/safety share at 1.000 (docs/method_review.md §4.2), and RE.2 reproduces that at every one of 378 cells. The repair is structural, not a reweighting: the catastrophe factors belong in the **admissibility of a policy** (a plan that collides is not a plan), and the comfort factor belongs in the **choice among admissible plans**. That is also how the dread zone and the comfort zone should differ formally.

4. Make the response model the switch, not a threshold. $P(\text{intervene})$ is the probability that the argmin is not "continue", marginalised over the belief. The per-driver random effect becomes θ_i rather than a level c_i , and the existing hierarchical estimator ($\text{fit_stage1_looming.fit_hier_lapse_gated}$) has to be re-pointed at that likelihood. The lapse stays; the "spread" does not — it is replaced by item 5.

5. Put the paradigm in the observation model, where it is already measured. The response spread σ_{resp} is not a driver property and should not be fitted per scenario. Card JJ.4 measured it as a property of the stimulus: $4.34 \times$

between the frozen clip and the car, and a sharper criterion for the larger object (truck σ_{resp} 0.273 against the car's 0.424, +60.6 held-out log-likelihood units under LOPO). Both are exactly what an observation model on (ϕ, φ) with angular noise predicts — apparent size enters the precision. **This is the one clean positive result of the whole JJ arc and it slots straight in.**

6. Include steering on the model side — this one is already demonstrated. Ruling JJ1.Q1 removed steering because the stimuli and the naturalistic data carry that constraint, which is right for the **response** and wrong for the **model**: the released planner escapes by steering in 378 of 378 of the study's own cells and never brakes. Card JJ.2b ran the correction and it moved the matched-TTC ordering from 0 of 24 to **19 of 24** rows in the human direction. A policy comparison that excludes the model's own chosen action is not a comparison, and the size of that movement is the best evidence in the whole arc that the framing — not the framework — is what has been wrong. Query RE1.Q2.

Note what this does *not* fix, because it is the hinge of the whole note: with the direction repaired, the held-out score is still 0.3161, chance. A right-signed but weak ordering is what you get when the quantity being ordered is dominated by a factor that points the other way. That is item 1.

7. Keep everything that is working. The cells, the folds, the weighted RMSE, the comparators on file, the pre-registration-before-the-run discipline, the worklog and the query register, and the rule that transfer is the test. None of that is implicated in any of the failures above.

5 The one thing that would make this fail, and it is the project's own central claim

The central claim is that active inference reduces the comfort-zone boundary to **one** scalar per driver, shared across scenarios. The reformulation makes that claim testable *for the first time*, because until now every test of it was confounded by the authors' calibration. But it can still fail, and it should be allowed to:

- A θ (or τ^{-1}) preference is the right comfort factor for **approach** conflicts. The left turn's axis is the oncoming vehicle's distance (card B.3.v2: 0.0558; looming ties at 0.0542 only because that design holds one speed per cell, so looming is unidentified there rather than rejected), and the cyclist overtake's is lateral clearance (card B.1). A single approach-rate factor will not carry the overtake, and the released lateral factor could not grade clearance at all (out/overtake_field_check.md: p +0.402 against the clearance label's -0.833).
- So the honest form of the claim is **two** comfort factors, not one: a longitudinal approach-rate factor and a lateral clearance factor, with the driver's parameters in both. If the two turn out to be independent across drivers, the "one scalar" claim is false and the elliptical joint-percentile fallback is what the deliverable becomes — which is the fallback already on file.

[Opinion: I think two factors is the likely outcome, and I think it is a better result than one, because it is what the scenario set actually supports and it still gives a single per-driver object — a point in a two-dimensional preference space — for the percentile.]

6 The order of tests, and the rules, pre-stated

Nothing below is authorized. Each is small, and each can kill the next.

RE.3 — is the comfort factor a comfort-zone boundary? Fit the three-parameter threshold model on the τ^{-1} factor's contribution and on a θ preference's contribution, on card JJ.2's own 378 cells, folds and metric. Rule: **the comfort factor earns the reformulation if it reaches within 0.01 of the gated looming rule's 0.1027 post-onset**, with the catastrophe factors excluded entirely. RE.2 already gives the τ^{-1} -inside-collision figure as 0.308, so this is a real test and it may well fail at the first step — the factor may have the right sign and the wrong shape.

RE.4 — does the switch reproduce the gate? With the catastrophe factors as admissibility and the comfort factor as the objective, compute $P(\text{argmin} \neq \text{continue})$ on the 90 pre-onset cells with nothing refitted. Rule: **below 0.05, card G.1's own criterion**, and with no gate term anywhere. This is the emergence claim, restated where it belongs.

RE.5 — one driver, two scenarios. Fit θ per driver on the cut-in; predict the left turn with nothing refitted. Rule: **within 0.01 of card B.3.v2's 0.0558**, and the per-driver Spearman inside card TR.1's interval [+0.407, +0.798]. This is the central claim.

RE.6 — the observation model, not a free spread. Refit RE.5 with σ fixed from card JJ.4's measured $4.34\times$ rather than free. Rule: **the held-out log-likelihood must not fall by more than 2 units**, the same margin JJ.4 used, or the precision story is decoration.

Only after RE.3 to RE.6 would anything be worth writing to the authors, and query RE1.Q1 asks Jonas about that separately.

7 What it costs

RE.3 is a day: the factor contributions already exist in `re2_preference_family.term_table`, and a θ preference is one function in `preferences.py` behind a flag defaulting off, with property tests. RE.4 is a day on top, because the admissibility form has to be written and the switch probability Monte-Carloed over the belief. RE.5 is the expensive one — a hierarchical fit whose per-driver parameter enters through a simulation rather than a closed form, which is days, not hours, and may need a different estimator (the current one integrates a one-dimensional level by Gauss-Hermite; θ entering through `argmin` is not that). RE.6 is hours once RE.5 exists.

[Opinion: RE.3 is worth doing regardless of what Jonas decides about the rest, because it is cheap and because a negative result there would close the whole line honestly. It is the single highest-value next card in the project right now.]

8 Queries

- **REF.Q1 (judgment, jonas):** the central proposal is to stop treating the braking margin as the comfort-zone boundary, which is the identification this project has been built on since gate R.1 and which is stated as such in `preferences.py` and in the handbook. RE.2 measures it at $p(\text{share}) = -0.861$. Adopt the separation (braking margin = dread boundary; an approach-rate factor = comfort boundary), or is there a reading in which the braking margin should still be the comfort boundary and the video paradigm is what is wrong?
- **REF.Q2 (judgment, jonas):** authorize RE.3 alone, the whole ladder RE.3 to RE.6, or none? RE.3 is a day and would close the line honestly if it fails.
- **REF.Q3 (judgment, review):** §5 argues the honest claim is two comfort factors, longitudinal and lateral, not one. That changes the deliverable from a percentile on one scalar to a joint percentile on two — which is the elliptical fallback already on file. Confirm that this is a permissible outcome rather than a failure of the project's premise.
- **REF.Q4 (minor, review):** a θ preference factor would be a genuine addition to the released $p(o)$, not a re-parameterisation. It must go behind a flag defaulting to the released behavior (standing rule 3) and it must never be called the authors' model in any document.
- **REF.Q5 (judgment, jonas):** if RE.3 succeeds, the result is that the published model contains a usable comfort-zone factor that its own magnitudes bury. That is a constructive finding about the paper and a natural thing to put to Julian in September. It interacts with RE1.Q1; both should be answered together.